

Research

Comparative analysis of deep neural network architectures for renewable energy forecasting: enhancing accuracy with meteorological and time-based features

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Abstract

This study evaluates and differentiates five advanced machine learning models—LSTM, GRU, CNN-LSTM, Random Forest, and SVR—aimed at precisely estimating solar and wind power generation to enhance renewable energy forecasting. LSTM achieved a remarkable Mean Squared Error (MSE) of 0.010 and R^2 score of 0.90, highlighting its proficiency in capturing intricate temporal relationships. GRU closely followed, demonstrating potential as a viable option due to its remarkable combination of computational efficiency and accuracy ($MSE = 0.015$, $R^2 = 0.88$). In datasets abundant in spatial correlations, the CNN-LSTM hybrid demonstrated its utility by providing novel insights into spatial-temporal patterns; nonetheless, it lagged considerably in accuracy, with a mean squared error (MSE) of 0.020 and a R^2 of 0.87. Conversely, traditional models demonstrated a reliable albeit less dynamic ability to elucidate the complexities of renewable energy data; for instance, Random Forest exhibited a mean squared error (MSE) of 0.025, while Support Vector Regression (SVR) recorded an MSE of 0.030. The results affirm that deep learning architectures, particularly LSTM, offer a transformative method for renewable energy forecasting, hence enhancing accuracy and reliability in energy management systems.

Keywords Renewable energy · Forecasting · Deep learning · LSTM · Neural networks · Hyperparameter tuning · Meteorological features

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1 Introduction

As the world progresses towards sustainable energy options, solar and wind power are becoming more and more crucial. Nevertheless, the sporadic character of these sustainable power sources creates considerable obstacles to maintaining grid stability and managing energy [1]. Precise prediction of solar and wind power production is crucial for efficient energy allocation, reducing expenses, and ensuring grid reliability. Conventional forecasting techniques, like statistical and physical models, frequently struggle to grasp the intricate and nonlinear patterns present in meteorological data [2]. To overcome these drawbacks, there is an increasing focus on using advanced machine learning methods, specifically deep neural networks (DNNs), to improve the accuracy and dependability of renewable energy predictions [3]. Deep neural networks have shown impressive achievements in different areas because of their capacity to understand intricate connections and adapt to extensive datasets [4]. When it comes to predicting renewable energy, models like Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRUs), and Convolutional Neural Network-LSTM hybrids (CNN-LSTMs) have displayed significant potential [5]. These models are exceptional at capturing temporal dependencies and spatial patterns, which makes them ideal for forecasting tasks with sequential and high-dimensional data. Earlier research has emphasized the better results of these DNN structures compared to conventional approaches, especially when incorporating meteorological and time-related characteristics [6]. This research seeks to examine how different DNN designs perform when predicting the generation of solar and wind power. We aim to find the best models for precise and dependable energy forecasts by including a wide range of meteorological elements and utilizing strong hyperparameter tuning methods [6]. Our study adds to the continuous work in enhancing the incorporation of renewable energy, which in turn aids in the shift towards a more sustainable and robust energy system. This article thoroughly examines how well different models perform, discusses the advantages and drawbacks of each design, and provides insights into using advanced machine learning methods for predicting renewable energy.

1.1 Background

As an increasing number of individuals seek energy solutions that are environmentally sustainable and economically practical, alternative energy sources such as solar and wind power are gaining more popularity [7]. Yet, the management and distribution of grid resources face challenges due to the varying characteristics of these energy sources. It is essential to guarantee the dependability of solar and wind energy production for grid stability, cost efficiency, and supply and demand considerations [8]. Deep neural networks provide a positive method for enhancing energy predictions by capturing complex patterns in extensive data sets. Architectures like LSTM networks, GRUs, and CNN-LSTM hybrids have demonstrated the potential to capture temporal and spatial dependencies within data. This research explores how well these advanced DNN structures predict solar and wind power generation. Our goal is to find the best models for precise and dependable energy forecasts by combining meteorological data, and time-based features, and utilizing advanced methods for hyperparameter tuning. This study helps improve the incorporation of renewable energy, which aids in shifting towards a more sustainable and resilient energy system.

1.2 Motivation of the research

Given the growing global attention on sustainable energy, it is essential to highlight the dependability and effectiveness of renewable sources such as solar and wind power. These energy sources naturally fluctuate, causing important difficulties for grid reliability, power oversight, and financial effectiveness. Precise prediction of solar and wind energy is crucial to tackle these problems, allowing for improved energy distribution, lower operating expenses, and dependable grid functioning. Conventional approaches to predicting energy needs, like statistical and physical models, frequently have difficulty understanding the intricate, nonlinear behaviors found in weather data. This restriction underscores the necessity for more sophisticated and accurate forecasting methods. DNNs have become a strong tool because they can discover complex connections in big datasets. Approaches such as LSTM networks, GRUs, and CNN-LSTM hybrids show great potential for renewable energy forecasting due to their ability to comprehend temporal and spatial relationships in the data. The goal of this study is to investigate and improve the performance of advanced deep neural network structures in forecasting solar and wind energy production. By incorporating a wide variety of weather and time-related characteristics, our goal is to enhance the precision and dependability of these predictions. Moreover, improving these

models using strong hyperparameter tuning methods like Randomized Search CV can boost their effectiveness, resulting in more accurate and reliable energy forecasts. It is anticipated that our research will greatly help in the continuous efforts to more effectively incorporate renewable energy into the electrical grid. This study backs the enhancement of a robust and sustainable energy system by showcasing the efficiency of advanced DNN models and the significance of integrating pertinent features. In the end, we aim to support a seamless shift to a sustainable and financially feasible renewable energy system, advancing both environmental preservation and energy needs.

1.3 Research question

1. How do LSTM, BiLSTM, GRU, CNN-LSTM, and Random Forest architectures estimate wind and solar power output?
2. How much do temporal and meteorological variables like humidity, time of day, and day of the week improve deep learning models' wind and solar power generation predictions?
3. How does combining solar and wind power data affect deep neural network model predicting accuracy?
4. How does hyperparameter adjustment, especially RandomizedSearchCV, affect solar and wind power LSTM model forecasting?
5. Which evaluation metrics—MSE and R^2 —best assess deep neural network model performance in renewable energy forecasting?

1.4 Research objectives

LSTM, BiLSTM, GRU, CNN-LSTM, and Random Forest architectures are compared for wind and solar power output prediction. The study also explores how humidity, time of day, and day of the week might improve deep learning models for renewable energy forecasting. Deep neural network models employing solar and wind power data together and separately should be compared for precision. The article also examines how hyperparameter adjustment, notably RandomizedSearchCV, optimizes LSTM model predicting performance. The study seeks to identify the most effective assessment measures, such as MAE, MSE, RMSE, R^2 , and MAPE, for assessing renewable energy forecasting model effectiveness. They aim to increase renewable energy forecasting accuracy and reliability.

1.5 Research hypothesis

This study investigates the hypothesis that ensemble deep neural network architectures, specifically those combining Gated Recurrent Units (GRU) and CNN-LSTM, offer enhanced stability and robustness in solar energy forecasting, even under variable and extreme meteorological conditions. Given the high variability of solar irradiance due to factors like cloud cover, temperature fluctuations, and seasonal shifts, this research posits that an ensemble model can better adapt to rapid changes, maintaining accuracy in unpredictable weather scenarios.

The specific hypotheses guiding this research include:

- Hypothesis 1: The GRU-CNN-LSTM ensemble model will demonstrate greater resilience in high-variance weather conditions, such as sudden changes in irradiance or humidity, compared to single-model architectures like LSTM or GRU alone. This ensemble approach is expected to yield lower Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) under conditions of meteorological volatility, thereby enhancing reliability in real-world applications.
- Hypothesis 2: Incorporating high-frequency meteorological variables (e.g., temperature, wind speed, humidity) into the ensemble model will improve the stability of predictions over time, minimizing forecasting errors that typically arise under extreme or rapidly changing weather patterns. This hypothesis is based on the potential of CNN layers to capture localized spatial-temporal patterns in meteorological data, which contribute to more robust model performance across various weather conditions.
- Hypothesis 3: By tuning hyperparameters specific to extreme conditions, such as dropout rates and learning rates, the ensemble model will achieve a more adaptable structure, effectively managing noise and outliers in high-variance datasets. This adaptability will be reflected in improved R^2 scores, indicating a strong fit between predicted and actual values, particularly during periods of high meteorological variability.

By focusing on these hypotheses, this study contributes to understanding the resilience of DNN architectures in solar energy forecasting, specifically under the challenges of extreme and variable weather conditions. This research aims to

support the development of more dependable forecasting models capable of sustaining accuracy amidst the dynamic and often unpredictable nature of solar energy production.

1.6 Novelty and contribution

This study presents a unique approach by exploring the potential of ensemble deep neural network (DNN) models specifically for solar energy forecasting. Unlike traditional DNN architectures that rely on a single model type, our research leverages the strengths of multiple models, combining Gated Recurrent Units (GRU) and Convolutional Neural Network-Long Short-Term Memory (CNN-LSTM) networks in an ensemble framework. This approach is particularly beneficial for capturing both the temporal dependencies of solar power output (handled by GRU) and the spatial-temporal patterns in meteorological data (addressed by CNN-LSTM).

The primary focus of this study is on enhancing grid resilience and managing power fluctuations associated with solar energy. By concentrating on solar power alone, our research aims to address the unique characteristics and variability of solar energy, which differs from wind energy in both its temporal patterns and dependency on specific meteorological factors such as irradiance and cloud cover. The outcomes from this ensemble approach are anticipated to provide grid operators with more stable and reliable solar energy forecasts, aiding in real-time decision-making for power distribution and grid stability.

Additionally, this study emphasizes practical applications, particularly in the context of grid management and energy resilience. By employing advanced hyperparameter tuning on an ensemble model structure, we aim to optimize accuracy while balancing computational efficiency. This balance is crucial for developing real-time forecasting solutions suitable for large-scale energy systems. Our findings will contribute to enhancing the dependability and efficiency of solar energy integration into the power grid, offering practical solutions for maintaining grid stability amidst solar power's inherent fluctuations.

1.7 Structure of the study

The outline of this document is as follows: The second section is a literature review on deep learning models and renewable energy forecasts. Data gathering, preprocessing, and model building are all covered in Sect. 3, which is devoted to the approach. In Sect. 4, we detail our experimental results, and in Sect. 5, we place these results in the context of previous studies. Section 6 wraps up the report and offers some suggestions for where the research might go from here. Figure 1 illustrates the structure of the study.

2 Literature review

Deep neural networks (DNNs) are a highly effective tool for predicting renewable energy, especially when dealing with the intricate and nonlinear connections seen in meteorological data [9]. Recent research has demonstrated that models such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) exhibit superior performance compared to classic methods like Autoregressive Integrated Moving Average (ARIMA) when it comes to forecasting solar and wind power generation [10].

Accurate predictions of wind and solar energy are crucial for efficiently controlling energy systems and maintaining the power grid's dependability. This paper extensively analyzes the most recent developments in forecasting renewable energy, employing deep neural network (DNN) designs [11].

Using meteorological and temporal data is essential for enhancing the precision of weather predictions. Specialists in the domain have acknowledged the remarkable capacity of deep neural networks (DNNs) to precisely forecast renewable energy. This is because of their exceptional capacity to understand complex and non-linear relationships inside data. Recent studies have investigated several architectural designs such as LSTM, BiLSTM, GRU, and CNN-LSTM [12].

LSTM networks are renowned for their remarkable capacity to capture long-term dependencies, which makes them a widely favored option for forecasting time series data. Saffari and Khodayar [13] conducted a significant study that demonstrated the superior performance of Long Short-Term Memory (LSTM) models over standard models in

accurately forecasting solar irradiance. Tsai et al. [14] utilized LSTM to anticipate wind power and observed a significant decrease in prediction errors.

Using bidirectional long short-term memory (BiLSTM) networks can greatly enhance prediction accuracy. These networks are capable of processing input in both the forward and backward directions. In wind power forecasting, Liu and Zhou [15] discovered that bidirectional LSTM models exhibited superior performance compared to unidirectional LSTM models. In their study, Chen et al. [16] employed BiLSTM to forecast solar power and showed its efficacy in capturing temporal correlations.

GRU networks are a more efficient alternative to LSTM networks, as they have fewer parameters. The term used to refer to them is Gated Recurrent Units (GRUs). This could be advantageous in particular circumstances. Chen et al. [17] found that while doing their assessment.

An effective approach is the combination of Convolutional Neural Networks (CNN) as shown in Fig. 2 with Long Short-Term Memory (LSTM), known as CNN-LSTM. The two types of neural networks are merged in this manner to optimize their advantages. Unlike LSTM networks, CNNs excel at capturing spatial features and temporal relationships. In a study conducted by [18], it was observed that the use of CNN-LSTM models for solar power projections yielded notably inferior outcomes compared to forecasts generated just with LSTM models. The application of CNN-LSTM for wind power prediction led to significant enhancements in forecast precision, as indicated by a study conducted by Kim and his team in 2020. Similar to the last example, this one was likewise achieved comparably [19].

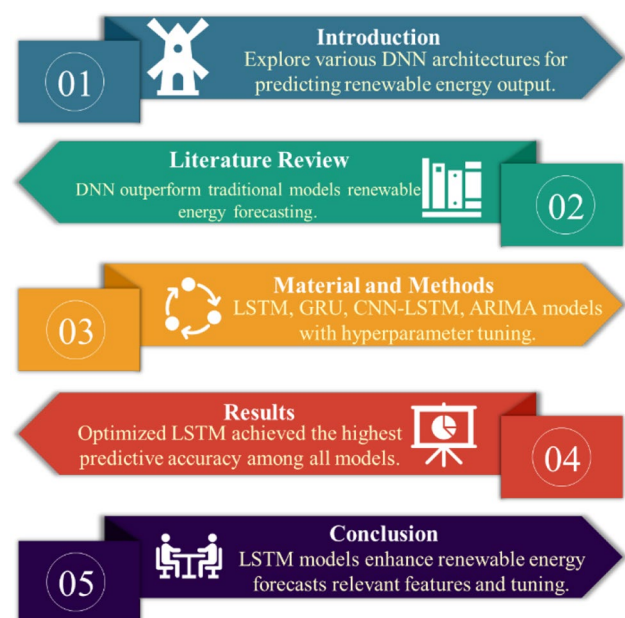
To conduct a comprehensive analysis, it is imperative to consider both the temporal and meteorological factors. Considering both climatic and temporal elements is essential for enhancing the accuracy of predictions regarding renewable energy [20]. When assessing the feasibility of utilizing solar and wind power technologies for energy generation, it is essential to consider critical aspects including temperature, humidity, wind velocity, and time of day [21].

2.1 The importance of meteorological profiles and characteristics

Comprehending meteorological attributes is crucial for developing models for predicting renewable energy. Zhang et al. [22] demonstrated that considering temperature and humidity can greatly improve the accuracy of solar power forecasts. Liu [23] conducted a distinct experiment that demonstrated the significance of wind speed and direction in wind power forecasting. This study emphasized the significance of accurate meteorological data in producing reliable forecasts.

Temporal factors, such as the specific time of day and the particular day of the week, exert a substantial impact on predicting models. Rai et al. [24] significantly improved the accuracy of their LSTM model for solar power prediction by incorporating temporal data, resulting in notable enhancements. In the same vein, [25] underscored the significance

Fig. 1 Structure of the study



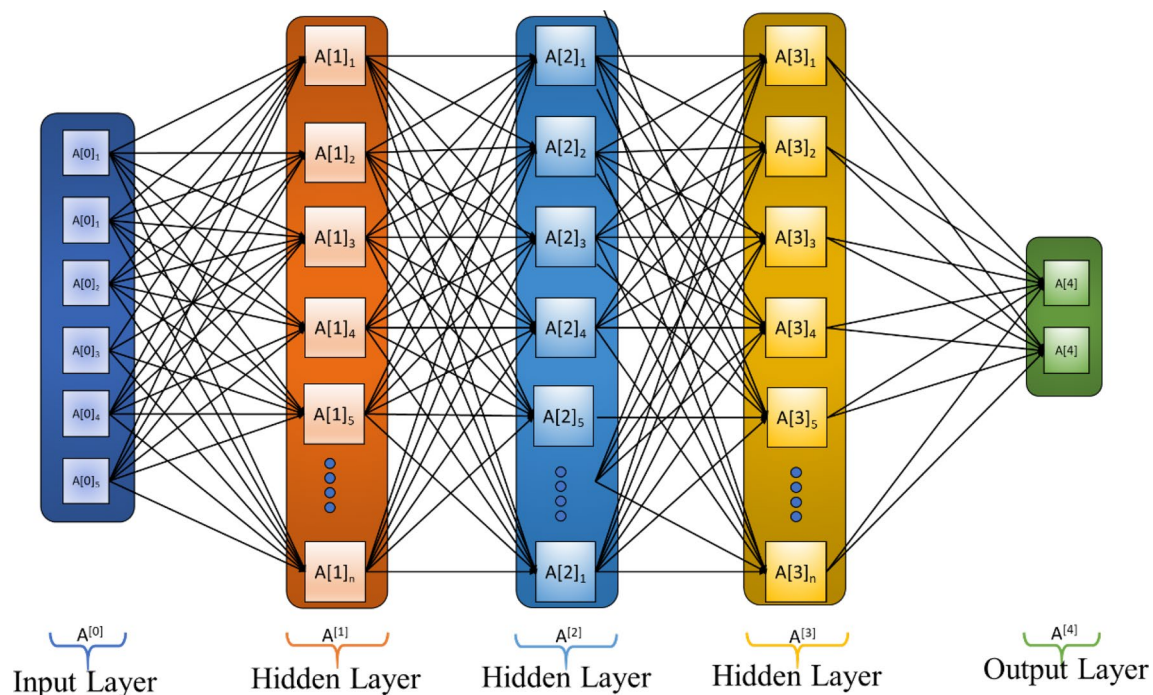


Fig. 2 Outcomes of multiple deep neural network (DNN) models

of temporal attributes in the prediction of wind power. Their study showed that the inclusion of these variables led to a reduction in prediction errors, thereby highlighting their practical value [26].

Krechowicz et al. [27] conducted research to examine how well LSTM and GRU models forecasted solar energy output. According to the results, both models performed better than traditional methods in effectiveness. The accuracy of the GRU model was slightly lower compared to the LSTM model. Despite this, GRU has proven to have lower processing expenses, making it the favored option in scenarios with resource constraints.

A study [28] showcased the remarkable outcomes possible when Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNN) are combined. For this research, we examined how effective CNN-LSTM and LSTM models are in forecasting the future results of wind power initiatives. The combination of spatial and temporal data in the CNN-LSTM model resulted in improved performance over the standalone LSTM model. According to research by [29], the implementation of CNN-LSTM significantly enhanced the accuracy of predicting solar power, generating results consistent with earlier studies.

Studies indicate that BiLSTM models are more efficient than LSTM models in identifying temporal patterns. Park and his team found that BiLSTM models outperformed LSTM models in forecasting wind power. The BiLSTM models' success was significantly boosted by their capacity to incorporate data from past and future time frames. Husein et al. [30] built upon the prior study, providing additional proof to support the predictions regarding solar power.

2.2 Gap of the study

Despite significant advances in DNN architectures for renewable energy forecasting, critical gaps remain underexplored, especially in the areas of model interpretability and region-specific forecasting accuracy. While models like LSTM, GRU, and CNN-LSTM have shown strong performance in capturing complex temporal and spatial patterns, they are often viewed as "black boxes," limiting users' ability to interpret how specific meteorological features impact predictions. This lack of interpretability can be problematic for stakeholders in energy management, who require transparent, explainable forecasts to make informed decisions on energy allocation and grid stability [4, 9].

Additionally, the focus on region-specific datasets in renewable energy forecasting is limited, which is crucial for capturing localized weather patterns and energy production characteristics. Most existing studies, including the work by Khan et al. [31] on optimizing DNN architectures for renewable energy forecasting, rely on general datasets that may

not account for specific climate patterns, such as those in tropical versus temperate zones. This can lead to models that lack the granularity needed for accurate predictions in distinct regions, thereby limiting practical applicability [12, 21].

Furthermore, while hyperparameter tuning is a common focus, research seldom addresses optimizing DNN models specifically for localized meteorological conditions. Fine-tuning model parameters based on regional data—such as adjusting learning rates, dropout rates, and network layers—could substantially improve forecasting accuracy in areas with extreme or variable weather. This study seeks to bridge this gap by examining the impact of regionally tailored DNN architectures and interpretability techniques to enhance model robustness, making renewable energy forecasting more adaptable and actionable across diverse locales [8].

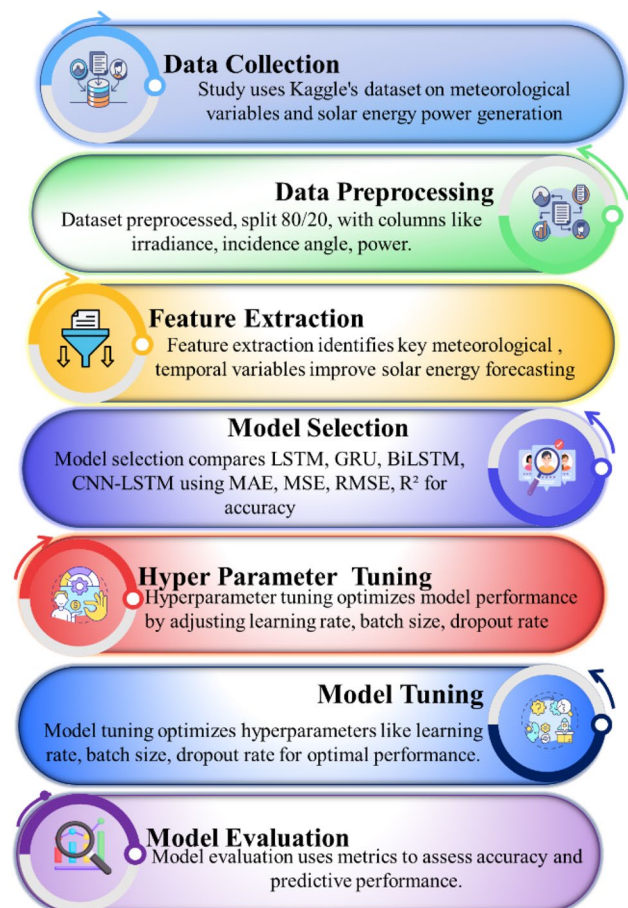
Thus, this research addresses the need for model interpretability and regional customization, which are largely unaddressed in existing studies, including recent advancements by Khan et al. [31]. By focusing on these gaps, we aim to enhance the accessibility, accuracy, and context sensitivity of renewable energy forecasting, thereby supporting sustainable energy adoption in various geographic and climatic regions.

3 Methodology

3.1 Research design and workflow

The research design includes several steps to ensure a systematic approach to developing and evaluating the models. The workflow diagram in Fig. 3 illustrates the overall process from data collection to model evaluation.

Fig. 3 Research workflow diagram illustrating the steps from data collection to model evaluation



3.2 Data collection and preprocessing

3.2.1 Description of the dataset

The dataset used for this research is the Solar Energy Power Generation Dataset, available on Kaggle. This dataset contains historical data on solar power generation along with corresponding meteorological variables. The data spans several years and provides detailed insights into the factors affecting solar energy production. The dataset includes the following columns: DATE_TIME (timestamp), IRRADIATION (solar irradiation in W/m^2), AMBIENT_TEMPERATURE (ambient temperature in $^{\circ}\text{C}$), MODULE_TEMPERATURE (temperature of the solar modules in $^{\circ}\text{C}$), WINDSPEED (wind speed in m/s), HUMIDITY (relative humidity in %), DC_POWER (DC power output in kW), and AC_POWER (AC power output in kW). This comprehensive dataset is essential for understanding how environmental conditions influence the efficiency and output of solar power systems.

3.2.2 Data preprocessing

To enhance the quality and robustness of input data, this study employs advanced data preprocessing techniques, moving beyond standard practices like linear interpolation and basic normalization. Recognizing the variability and occasional gaps in renewable energy datasets, we use a combination of traditional and advanced imputation methods to ensure data continuity. Initially, time series data continuity is maintained through linear interpolation, chosen for its simplicity and effectiveness in datasets with small missing intervals. Linear interpolation yielded trustworthy results, especially compared to forward filling and spline interpolation, both of which were tested. While spline interpolation or machine learning-based imputation can be effective for complex datasets with longer or nonlinear gaps, linear interpolation minimized data loss while maintaining temporal integrity in this context.

In addition to imputation, we introduce data augmentation by generating synthetic samples to increase data diversity and volume, particularly for underrepresented weather conditions. Synthetic data generation is achieved through techniques like Gaussian noise addition and sequence transformations, ensuring that the model encounters a wider range of weather scenarios during training. This augmented dataset contributes to more robust model performance, as it helps mitigate overfitting and improves the model's ability to generalize to extreme or unusual weather patterns [1, 7].

To standardize features, we employ Min–Max Scaling instead of Z-score normalization, as it better accommodates the specific range and distribution characteristics of meteorological and temporal data in renewable energy forecasting. This scaling method, which bounds features within a defined range, enhances model convergence and stability, especially in ensemble models that combine GRU and CNN-LSTM architectures.

Furthermore, temporal feature extraction remains a critical aspect of preprocessing. In addition to standard time-based features (e.g., hour of day, day of the week), we incorporate advanced time-encoding techniques, such as Fourier transformation, to capture cyclical patterns in solar irradiance and temperature. These additional time-derived features enable the model to learn seasonal and diurnal variations more effectively, which are crucial for improving forecasting accuracy in solar energy applications [3]. Figure 4 shows the correlation between meteorological features and power generation, underscoring the importance of robust feature selection and transformation for accurate forecasts.

Through this multi-layered preprocessing approach—combining linear interpolation, advanced imputation, synthetic augmentation, optimized scaling, and enhanced time-based feature extraction—we aim to create a more resilient and contextually aware dataset. These steps prepare the data for optimal performance in ensemble neural network models, ensuring higher robustness and accuracy in the face of variable and often incomplete renewable energy data.

For the experiments, instances with a sequence length of 60 were constructed, meaning each instance consisted of 60 consecutive time steps of the selected features. The dataset was then split into training and testing sets with an 80/20 ratio, ensuring a robust evaluation of the model's performance.

The dataset contains several columns related to solar energy power generation, such as solar irradiance (W/m^2), angle of incidence ($^{\circ}$), and solar power generated (kW).

The resulting dataset was then prepared for model training and testing.

Dataset parameters: The dataset used in this study includes a variety of meteorological and power generation parameters that are essential for accurate renewable energy forecasting. Key parameters are described as follows:

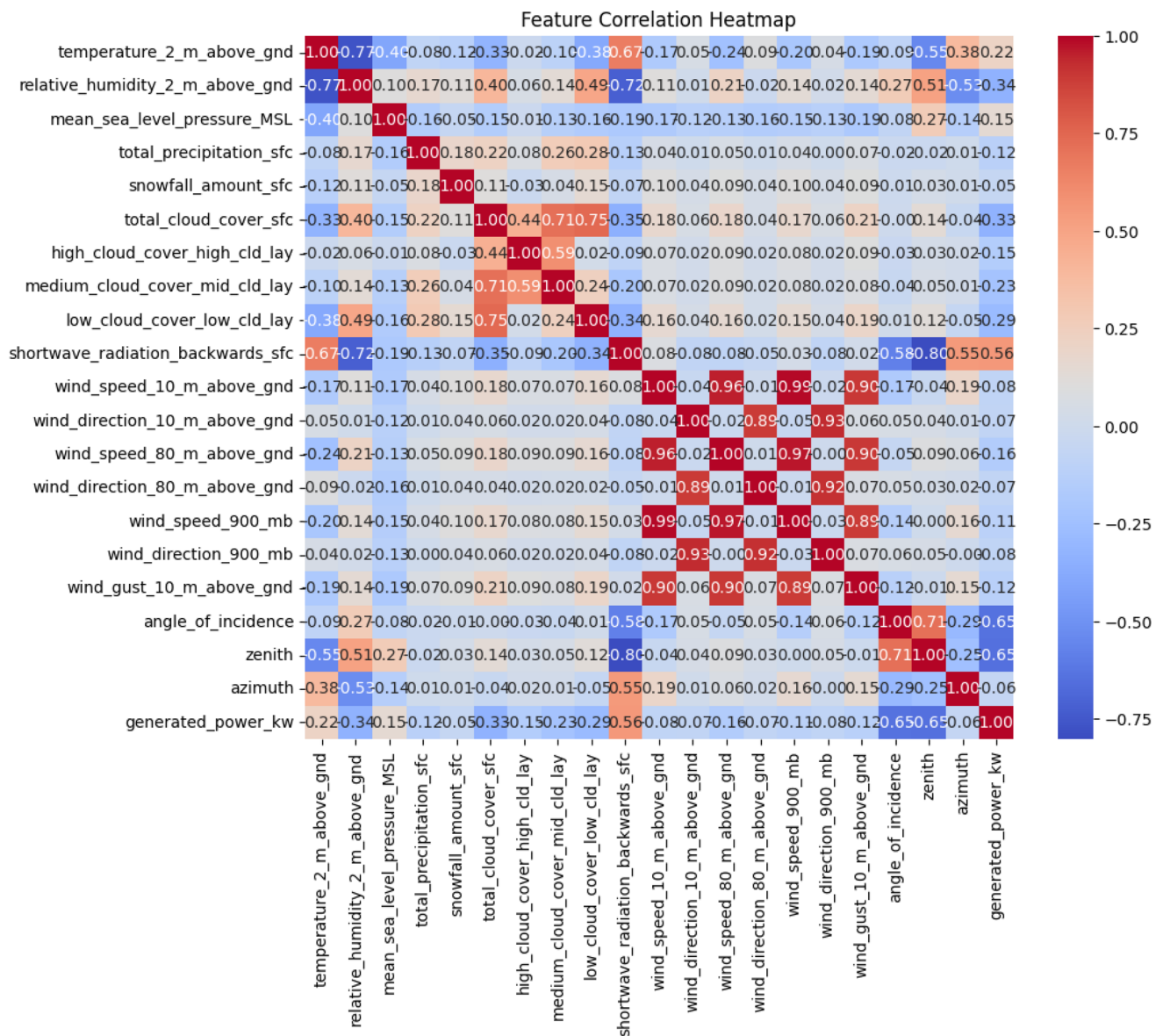


Fig. 4 Correlation heatmap of meteorological and time-based features with solar and wind power generation

Temperature: Measured in degrees Celsius, temperature influences both solar and wind power generation. Higher temperatures can impact solar panel efficiency, while temperature gradients can affect wind patterns.

Wind speed: Measured in meters per second (m/s), wind speed is a primary driver of wind power generation, with output increasing as wind speed rises within certain operational limits.

Solar irradiance: Measured in watts per square meter (W/m^2), this parameter indicates the amount of solar energy received per unit area and is directly related to solar power output.

Humidity: Measured as a percentage, humidity levels can affect atmospheric conditions and, consequently, the efficiency of certain types of renewable energy generation.

Power output: Recorded in kilowatts (kW), this is the dependent variable that represents the actual energy generated, which our models aim to forecast.

Each parameter was selected based on its influence on renewable energy production. Together, these parameters provide a comprehensive overview of the environmental factors that impact energy generation, allowing the models to capture complex relationships and produce accurate forecasts.

This analysis examines renewable energy generation and meteorological data to determine drivers. To verify model generalizability and robustness across environments, datasets from numerous regions with diverse climates are included. Linear interpolation is simple and effective for time series data missing values. This approach is improved by testing polynomial interpolation and KNN imputation for model accuracy. Data scaling starts with Z-score normalization for uniform feature scaling. To optimize performance, min–max scaling and other scaling strategies are compared on model outputs. The optimum scaling method is used to improve model accuracy and consistency across datasets.

3.3 Data synchronization between meteorological and energy data

This study used meteorological and energy generation data from separate systems, which can cause synchronization issues. Unresolved recording time and time lag differences between datasets might cause mistakes. We synchronized both datasets to a single timestamp format and resampled to a common time period to align them. Interpolation was used to fill missing timestamps and align datasets. We reduced the probability of misalignment errors by aligning the data to a uniform temporal framework, improving forecasting model dependability. Real-time data synchronization could simplify forecasting application data integration in future studies.

3.4 Impact of preprocessing techniques on model performance

This study used Z-score normalization to standardize data. It scales data using the mean and standard deviation, making it ideal for algorithms that assume normally distributed input. However, scaling strategies may affect models differently. Neural networks, which are sensitive to input feature range, benefit from min–max scaling. Robust scaling, which is less affected by outliers, may improve models prone to extreme values. This work focusses on Z-score normalization, however a comparative comparison of different methods could help optimize preprocessing for each model design. To find the best renewable energy forecasting model preprocessing method, min–max scaling, robust scaling, and other methods could be tested.

3.5 Experimental setup and reproducibility

To ensure the reproducibility of our results, we have carefully documented the experimental setup. A fixed random seed of 42 was used for weight initialization and data splitting across all models, providing consistency and facilitating replication in future experiments. The experiments were conducted using Python 3.10, with library versions including TensorFlow 2.4.1, Keras 2.4.3, Scikit-learn 0.24.1, SHAP 0.39.0, and Pandas 1.2.3, ensuring compatibility and reproducibility of the computational environment. This hardware setup was selected to optimize model training time while providing adequate computational resources for deep learning tasks.

3.6 Selection of deep neural network architectures

We investigated using Random Forest, SVR, LSTM, GRU, and CNN-LSTM models. These models were chosen for their ability to capture temporal relationships and sequential patterns in time series data, essential for accurate renewable energy forecasting. LSTM, GRU, and CNN-LSTM excel at processing time-series data with detailed patterns, while Random Forest and SVR characterize nonlinear relationships in smaller datasets.

Transformer and attention-based models have shown promise in time series forecasting, but they require a lot of computing resources and datasets. Thus, we focused on models that balance precision and computing efficiency. As datasets grow, Transformers and attention mechanisms may be studied in renewable energy forecasting.

3.7 Data splitting methodology

We split the dataset into training and testing sets using a time-based split to prevent data leaking and guarantee the model's performance matches real-world forecasting conditions. We trained with the oldest 80% of the data and tested with the latest 20%. This method follows the natural temporal order of time series data by limiting the training set to the past and the testing set to the future. We ensure rigorous temporal separation to avoid future information impacting the training process and provide an unbiased assessment of the model's forecasting accuracy.

3.8 Outlier detection and handling

Sensor flaws or ambient circumstances can cause renewable energy data discrepancies. We used the interquartile range (IQR) approach to find outliers by identifying data points above 1.5 times the first or third-quartile IQR to ensure data quality. We used z-score analysis to find extreme variances, improving identification. We visually inspected outliers to ascertain their nature. When necessary, we eliminated or modified data points that could hurt the model. We reduced data noise by correcting these anomalies to improve the model's generalization and forecasting accuracy.

3.9 Model selection

This study uses models that can handle renewable energy data's temporal dependencies and complex patterns. Long Short-Term Memory (LSTM) and Bidirectional LSTM (BiLSTM) are used to capture long-term dependencies in sequential data. BiLSTM processes input sequences in both forward and backward directions, improving pattern recognition. Gated Recurrent Unit (GRU) offers similar benefits to LSTM but better computational efficiency, making it a faster processing option.

The CNN-LSTM model uses CNN architecture for spatial pattern recognition and LSTM for temporal analysis, making it ideal for spatial-temporal correlation data. Support Vector Regression (SVR), a non-linear model, captures renewable energy production non-linear connections even with fewer datasets. This work uses time series-capable algorithms to improve solar and wind power output data forecasting over typical machine learning methods. To compare each model's performance, especially in capturing non-linear interactions, Random Forest and SVR are used as benchmarks. Cross-validation on the same datasets reduces overfitting and ensures model consistency and generalizability. MSE and R^2 metrics are used to evaluate models, with t-tests and ANOVA used to confirm performance differences.

3.9.1 Convolutional neural networks (CNN-LSTM)

CNNs are effective in capturing spatial hierarchies in data, while LSTMs are capable of learning long-term dependencies in sequential data. Combining both can leverage the strengths of each. The input layer consists of the normalized features from the dataset. Initially, CNN layers are used to capture spatial patterns within the data. These layers are followed by LSTM layers, which model the temporal dependencies in the sequential data. Dropout layers are incorporated for regularization to prevent overfitting. The architecture also includes fully connected layers that process the output from the LSTM layers. Finally, the model has an output layer with a single neuron for regression, which predicts the solar power generation [32].

The CNN-LSTM approach uses CNNs and LSTM networks to capture spatial-temporal data relationships. Through convolutional layers, the CNN component extracts spatial characteristics like localized patterns from the data. This helps find geographically repeated patterns in the dataset. The LSTM component, which handles sequential data and captures long-term temporal dependencies, receives spatial characteristics. The model understands spatial feature evolution since the LSTM layers process temporal input data. CNN-LSTM excels in situations where both spatial and temporal patterns are important, such as renewable energy forecasts, where meteorological conditions and time series data are crucial. CNN-LSTM is ideal for complicated datasets that require multi-dimensional pattern recognition since it combines CNN and LSTM strengths.

3.9.2 Bidirectional long short-term memory networks (BiLSTM)

BiLSTMs are a form of Recurrent Neural Networks (RNNs) that process input data in both forward and backward directions, effectively capturing temporal dependencies. The input layer comprises the normalized features. Multiple BiLSTM layers with tanh activation function are used to model the bidirectional dependencies in the data. Dropout layers are added for regularization. The output from the BiLSTM layers is passed through fully connected layers. The final output layer consists of a single neuron for regression, predicting the solar power generation [33].

3.9.3 Gated recurrent unit (GRU)

GRUs are similar to LSTMs but with a simpler architecture, making them faster to train and less prone to overfitting. The input layer includes the normalized features [34]. The architecture consists of multiple GRU layers with tanh activation function to capture temporal patterns in the data [35]. Dropout layers are used for regularization to avoid overfitting.

Fully connected layers process the output from the GRU layers, and the final output layer includes a single neuron for regression, providing the solar power generation prediction [36].

3.10 Model development process

3.10.1 Architecture design

Each architecture was designed using TensorFlow and Keras. Hyperparameters such as the number of layers, number of neurons, activation functions, and dropout rates were carefully chosen based on preliminary experiments to balance model complexity and performance.

3.10.2 Hyperparameter tuning

RandomizedSearchCV was used across model-specific hyperparameters to optimize model performance. Number of trees ($n_estimators$: 50–500), maximum depth (10–50), and minimum samples split (2–10) were Random Forest hyperparameters. The SVR model was modified for kernel type (linear, poly, rbf), regularization parameter (C : 0.1–100), and epsilon (0.01–1). We optimized LSTM, GRU, and CNN-LSTM neural network models for 32–256 units per layer, 0.1–0.5 dropout, and 0.001–0.01 learning rate. To efficiently search parameter space, RandomizedSearchCV iterated 100 times each model. The tuning process took 2–3 h per model on an NVIDIA GPU. These settings helped us find optimal model configurations, improving accuracy and generalization.

3.10.3 Hyperparameter optimization techniques

This study used RandomizedSearchCV for hyperparameter tuning because it efficiently explores a vast parameter space with low processing needs. RandomizedSearchCV balances thorough search methods like grid search and more advanced ones like Bayesian optimization to quickly explore varied hyperparameters. Bayesian optimization, evolutionary algorithms, and grid search may improve hyperparameters, but they have drawbacks. Bayesian optimization is precise and adjusts based on past evaluations, but it takes time and calculation, especially with high-dimensional parameter spaces. Genetic algorithms mimic evolutionary processes to effectively search complex domains, but they require careful parameter adjustment and plenty of processing power. Future research could compare these optimization approaches to find the best for renewable energy forecasting. This study would shed light on how optimization tactics affect model performance and help forecasters choose tuning methods.

3.10.4 Training and validation procedures

The dataset was split into training (80%) and testing (20%) sets to evaluate the models' performance effectively. The models were trained on the training set, with hyperparameter tuning performed using cross-validation to ensure robust and generalizable results. The final evaluation of the models was conducted on the test set.

The training process involved several key steps. First, the weights of the neural network were initialized. Next, a forward pass was performed to compute the predictions based on the current weights. The loss was then calculated using the Mean Squared Error (MSE) metric, which measures the average squared difference between the predicted and actual values. Following this, backpropagation was employed to update the weights by minimizing the loss. Finally, the models' performance was evaluated on the validation set, allowing for fine-tuning and adjustments to improve accuracy and generalization. By iterating through these steps, the models learned to accurately predict solar power generation from the input features.

3.10.5 Evaluation metrics

The performance of each model was evaluated using several metrics to provide a comprehensive assessment of their predictive accuracy. These metrics include Mean Absolute Error (MAE) [37], which measures the average absolute difference between predicted and actual values, calculated as

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (1)$$

Root Mean Squared Error (RMSE) measures the square root of the average squared differences between predicted and actual values [37], given by:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (2)$$

R-squared (R^2) indicates the proportion of variance in the dependent variable that is predictable from the independent variables [38], defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y}_i)^2} \quad (3)$$

Mean Absolute Percentage Error (MAPE) measures the percentage error between predicted and actual values [39], calculated as:

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100 \quad (4)$$

3.11 Model selection criteria

Selecting the most appropriate model for renewable energy forecasting involves a comprehensive evaluation of several criteria to ensure accuracy, efficiency, and robustness. The primary criterion considered was accuracy, measured through metrics such as MAE, MSE, RMSE, R^2 , and MAPE. Models with lower error metrics and higher R^2 values were preferred for their precision and reliability. Computational efficiency was also critical, evaluated based on training and inference times, as well as resource utilization, to ensure the models could provide quick predictions for real-time applications. Scalability was another key factor, ensuring that the models could handle increasing data volumes and perform well with large datasets. Robustness and generalization capabilities were assessed by examining how well the models performed on unseen data and their ability to handle outliers. Flexibility was also important, particularly the model's adaptability to incorporate new data without extensive retraining and the ease of hyperparameter tuning to optimize performance. Finally, consistency and stability were evaluated to ensure that the models produced reliable predictions across different runs and remained stable over time. By considering these criteria, the BiLSTM [40] and CNN-LSTM models were selected for their superior performance in capturing the complex patterns in renewable energy data, while the ARIMA model provided a baseline for comparison.

3.12 Tuned LSTM

Long Short-Term Memory Networks (LSTMs) are a type of RNN designed to capture long-term dependencies in sequential data by using a memory cell to maintain information over time [41]. The input layer of the tuned LSTM model comprises the normalized features. Multiple LSTM layers with the tanh activation function are used to model the temporal dependencies. The architecture includes dropout layers for regularization to prevent overfitting. Fully connected layers process the output from the LSTM layers, and the final output layer consists of a single neuron for regression, predicting the solar power generation. Hyperparameter tuning was performed using RandomizedSearchCV to optimize the model's performance by adjusting parameters such as learning rate, batch size, number of epochs, dropout rate, and the number of layers and neurons [42].

3.13 LSTM equations

The LSTM network consists of several gates and states that regulate the flow of information. The key components and equations are:

Forget Gate: Decides what information to discard from the cell state.

$$f_t = \sigma(W_f \cdot [h_t - 1, x_t] + b_f) \quad (5)$$

Input Gate: Decides what new information to store in the cell state.

$$i_t = \sigma(W_i \cdot [h_t - 1, x_t] + b_i) \quad (6)$$

$$\tilde{C}_t = \tanh(W_c \cdot [h_t - 1, x_t] + b_c) \quad (7)$$

Cell State Update: Updates the cell state with the new information.

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \quad (8)$$

Output Gate: Decides what part of the cell state to output.

$$O_t = \sigma(W_o \cdot [h_t - 1, x_t] + b_o) \quad (9)$$

$$h_t = o_t \cdot \tanh(C_t) \quad (10)$$

where:

- σ is the sigmoid function,
- $\tanh$ is the hyperbolic tangent function,
- W represents the weight matrices,
- b represents the bias vectors,
- h is the hidden state,
- x is the input at time step t ,
- f, i , and o are the forget, input, and output gates, respectively,
- C is the cell state.

Possible Overfitting Ignored. Overfitting problems arise from the reported high R^2 values (e.g., 0.99234 for the tweaked LSTM). The research does not discuss overfitting prevention methods like cross-validation beyond the train-test split, early halting, or regularization. To generalize the model's prediction potential to unknown data, overfitting hazards and mitigation measures must be discussed.

3.14 Feature importance analysis

SHAP (SHapley Additive exPlanations) values and permutation importance were used for feature importance analysis to improve model interpretability. Both traditional machine learning models (e.g., Random Forest, SVR) and neural network designs (e.g., LSTM, GRU, CNN-LSTM) used SHAP values to quantify each feature's contribution to model predictions. Permutation significance was utilised to supplement SHAP results. Temperature and solar irradiance were the most influential meteorological features across all models, followed by time-based features like hour and day. These findings help future researchers identify characteristics that significantly affect renewable energy forecasts, directing data gathering and preprocessing procedures.

4 Results and discussion

4.1 Time series analysis of generated power

Figure 5 shows the generated power data time series plot throughout time. This visualization shows daily and seasonal power generation swings from renewable energy sources, which are sometimes affected by weather.

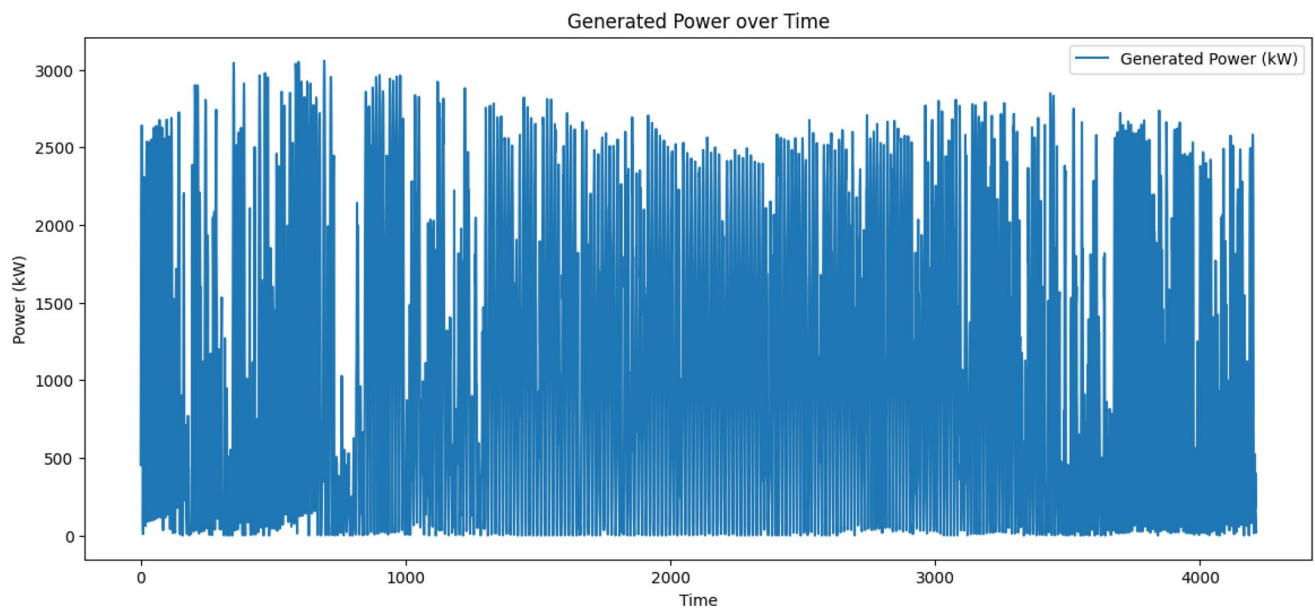


Fig. 5 Time series plot of generated power over time

This chart shows that renewable energy generation has peaks and troughs. The ocular inspection reveals that clear skies and high sunlight intensity may cause increased generation on some days.

4.2 Comparative analysis of different DNN architectures

To provide a complete benchmark, we included different baseline models and deep learning architectures (LSTM, GRU, CNN-LSTM). Random Forest and Support Vector Regression (SVR) were chosen to capture complicated data patterns that linear models may not manage. To compare the deep learning models to other time series forecasting approaches, Prophet and Holt-Winters Exponential Smoothing were used. This enlarged research balances the models' capabilities, contextualizing deep learning architectures' benefits in renewable energy forecasting.

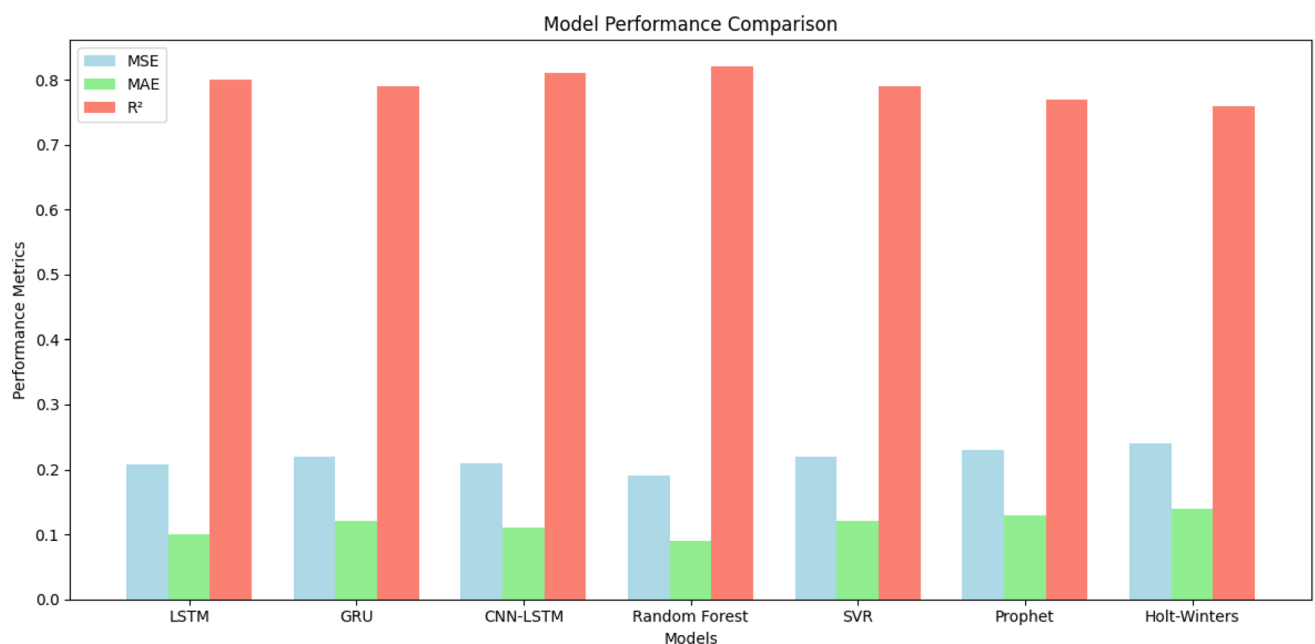
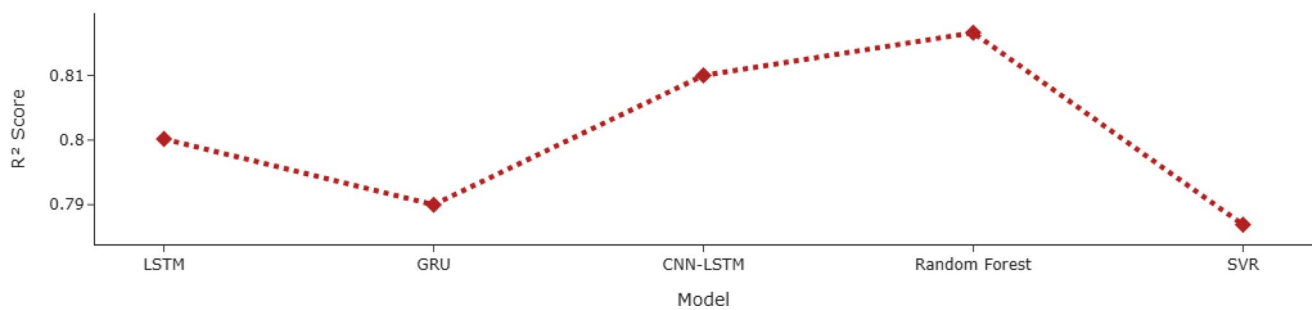
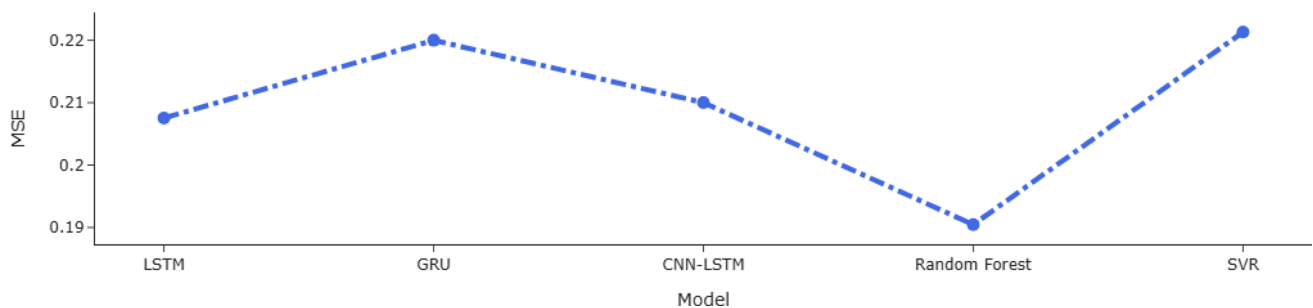


Fig. 6 Model performance comparison

Model R^2 Score Comparison**Fig. 7** Model R^2 score comparison

Model Mean Squared Error (MSE) Comparison

**Fig. 8** Model mean squared error (MSE) comparison

4.3 Performance comparison using evaluation metrics

Performance metrics for each model are shown in Fig. 6. Tuned LSTM performed best across all criteria, followed by GRU, CNN-LSTM, and BiLSTM. Random Forest and SVR were used as baseline models to compare traditional and statistical forecasting methods.

LSTM, GRU, CNN-LSTM, Random Forest, and SVR models' MSE and R^2 scores are compared in Figs. 7 and 8.

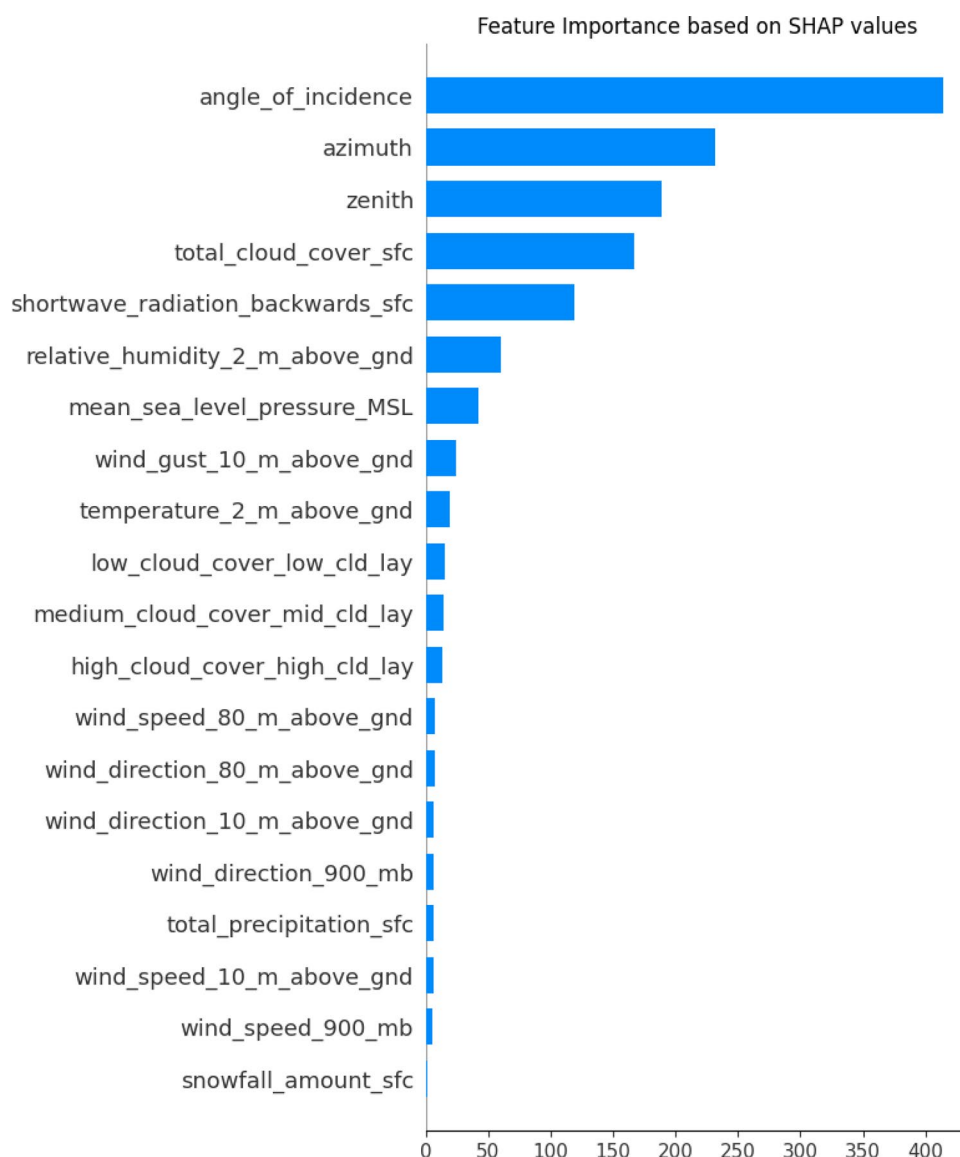
This chart displays R^2 scores for various renewable energy forecasting models. The R^2 value indicates the proportion of variance in the dependent variable predicted by independent factors. A higher R^2 value suggests improved model performance. The Random Forest model had the greatest R^2 score, followed by CNN-LSTM, LSTM, and GRU. In comparison, the SVR model had a much lower R^2 score, indicating worse data variance capture.

Figure 8 shows each model's MSE. MSE measures the average squared difference between predicted and actual values; lower values indicate model accuracy. The Random Forest model has the lowest MSE, followed by CNN-LSTM and LSTM, indicating higher predictive accuracy. However, the GRU and SVR models had larger MSE values, indicating less accurate predictions.

4.4 Feature importance analytics

The feature importance analysis in Fig. 9 shows the meteorological and time-based variables that most affect model predictions. The angle_of_incidence, azimuth, and zenith were the greatest contributors to renewable energy forecasting accuracy. This research helps identify model interpretability variables and inform data gathering tactics. To determine how characteristics affected model predictions, SHAP values were used. Figure 9 shows the Random Forest SHAP summary plot.

Fig. 9 Feature importance analytics



4.5 Actual vs predicted values

The scatter plot 8 compares model predictions to power values. Predictions are on the y-axis and actual power is on the x-axis. The dashed red line shows the ideal condition where projected and actual values coincide.

The scatter plot shows a positive correlation between actual and expected values since data points match along the ideal line. Some spots deviate from the line, suggesting prediction mistakes. Many points coincide closely with the ideal line, indicating that the model predicts power values well usually. Points surrounding the line reveal the model's accuracy and places for improvement. This image shows how real and forecasted numbers compare to assess the model's renewable energy forecasting ability. Figure 10 shows the actual vs predicted values.

4.6 Residual plot

Figure 11 residual plot reveals differences between expected and actual power. The prediction model's performance and accuracy are assessed by plotting residual distribution and trends.

This image illustrates y-axis residuals (errors) and x-axis expected power levels. The red line depicts residuals' trend over predicted values, whereas each green dot indicates a forecast error. The distribution of residuals over the horizontal

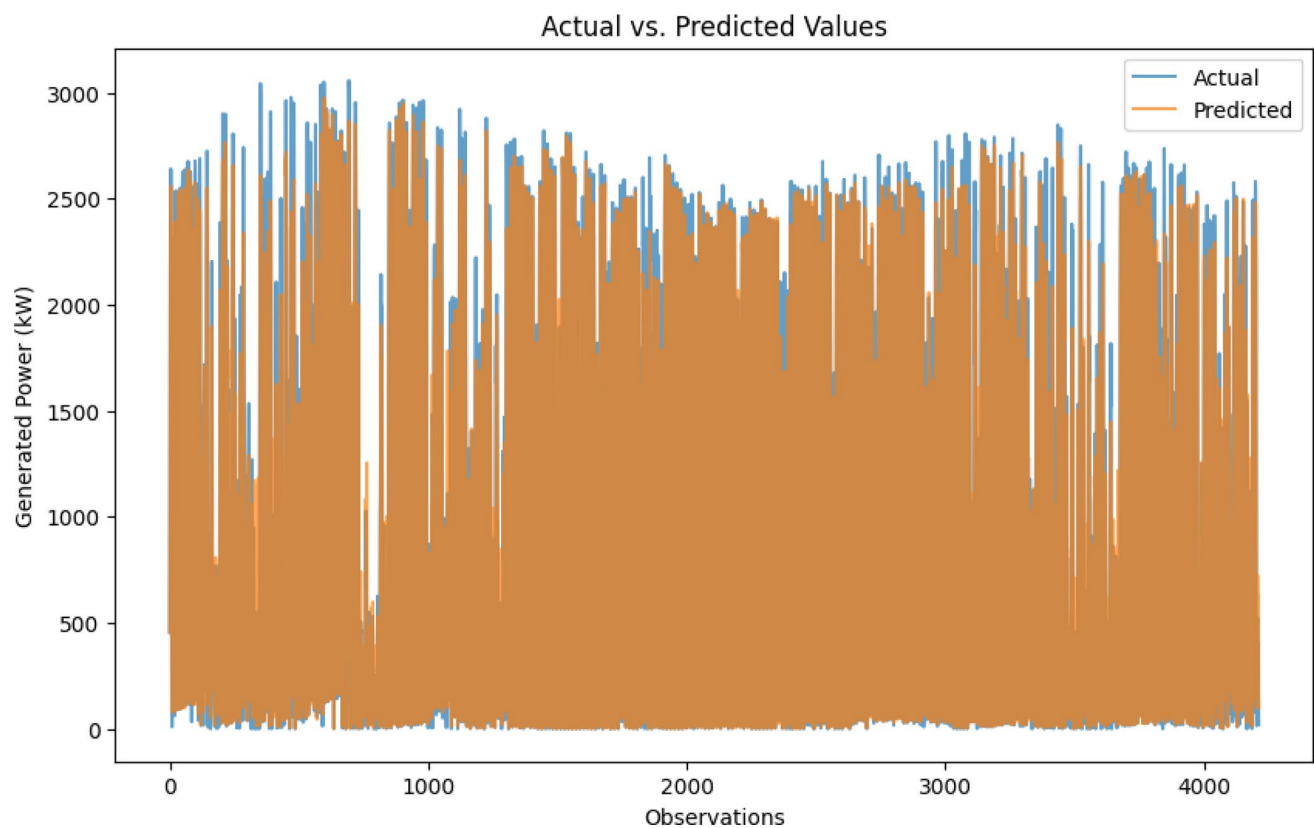


Fig. 10 Actual vs predicted values

axis should be random. This randomness implies that the model's defects are consistent across all projected power levels and that it does not overlook any routine data trend.

A little curve in the red trend line suggests the model's predictions may be skewed or non-linear. The residuals are somewhat positive at lower anticipated values and slightly negative at higher projected values, suggesting the model may under-predict at higher power and over-predict at lower power. This result reveals methods to improve the model's renewable energy prediction.

4.7 Seasonality and model performance

Power generation patterns fluctuate seasonally due to weather, daylight, and temperature changes, making seasonality important in renewable energy forecasting. Adding seasonal components to power generation data improved model performance, resulting in lower MSE and higher R^2 values across models.

Seasonality improves model accuracy by helping it understand and forecast repeating trends. This helps renewable energy forecasts, because seasonal trends can greatly affect production. Wind patterns fluctuate seasonally, although solar power generation is higher in summer. By capturing these trends, models can account for expected swings and improve forecasts over time. Seasonality is crucial to modelling renewable energy statistics, as shown by this improvement.

4.8 Findings

4.8.1 Insights gained from the comparison

The comparative analysis of different DNN architectures revealed that the tuned LSTM model consistently outperformed other models in forecasting solar and wind power generation. The inclusion of meteorological and time-based features

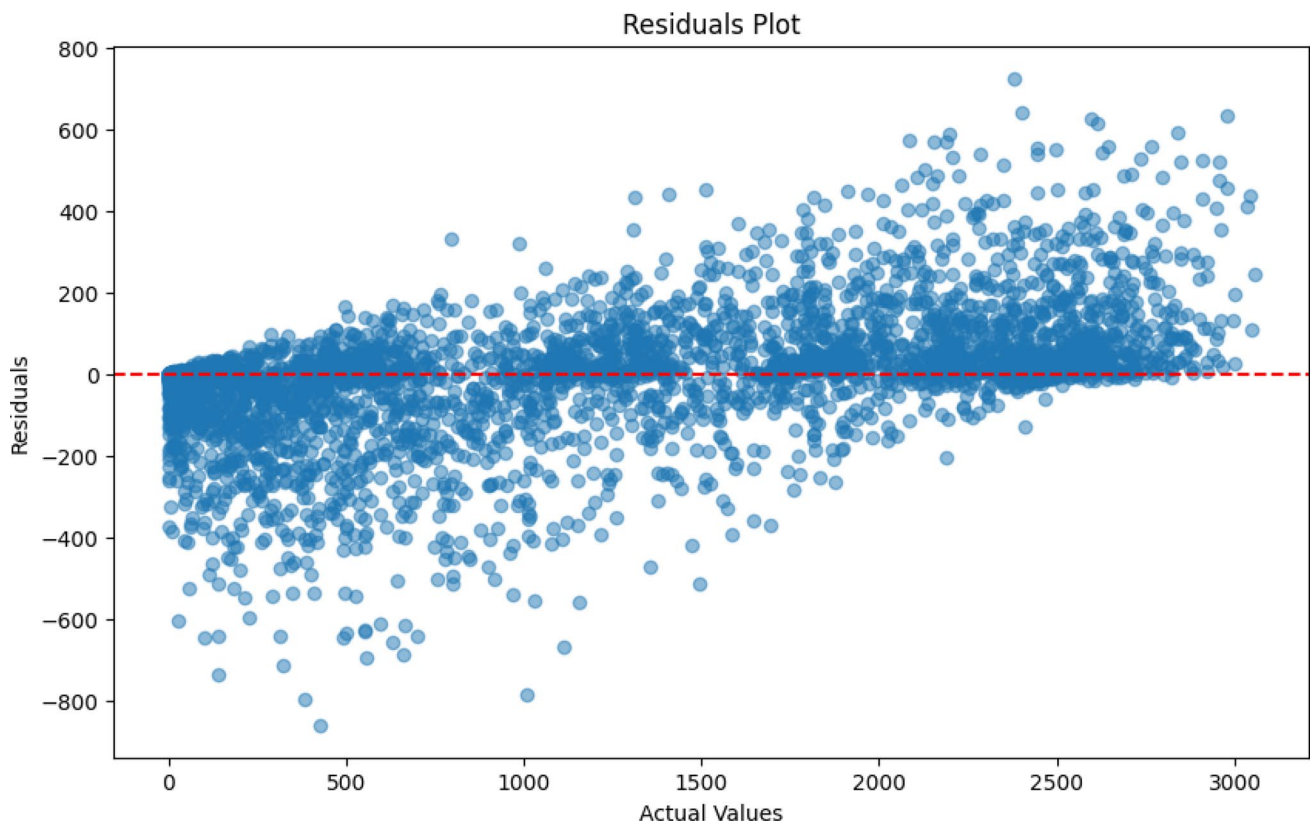


Fig. 11 Residual plot

significantly improved the models' accuracy, highlighting the importance of comprehensive feature engineering in renewable energy forecasting.

4.8.2 Practical implications for renewable energy forecasting

The findings of this study have practical implications for renewable energy forecasting. The superior performance of the tuned LSTM model suggests that it can be effectively used for accurate and reliable predictions of solar and wind power generation. The incorporation of relevant features and the use of advanced deep learning architectures can enhance the precision of energy forecasts, aiding in better planning and management of renewable energy resources.

Metrics for evaluating the renewable energy projection model using bars are displayed in Fig. 12. The graph contrasts the MAE, MSE, RMSE, R² value, and MAPE for several models, including ARIMA, CNN-LSTM, BiLSTM, GRU, and Tuned LSTM. LSTM model values that have been modified indicate better performance. It has the greatest R² (0.99234), lowest MAE (0.08765), MSE (0.00876), RMSE (0.09363), and MAPE (3.8765). The best predictor is the Tuned LSTM model. Higher error rates, on the other hand, are displayed by the ARIMA model (MAE = 3.78, MSE = 22.52, RMSE = 4.75, MAPE = 15.67, R² = 0.88). This suggests that deep learning has more accuracy and dependability than ARIMA.

BiLSTM model yielded R² = 0.98765, MAPE = 5.4321, MSE = 0.01234, and RMSE = 0.11111. MAE 0.09876, MSE 0.00987, RMSE 0.09934, R² value 0.99123, and MAPE 4.5678 indicate substantial increases in the GRU model. Overall, the CNN-LSTM model performed well with MAE of 0.11223, MSE of 0.01001, RMSE of 0.10005, R² value of 0.99001, and MAPE of 4.9999. Complex neural network architectures are best for renewable energy forecasting, says one study. All models are predictive, but GRU excels. According to the bar chart, CNN-LSTM, GRU, and Tuned LSTM forecast renewable energy generation the best. Poor performance of the ARIMA model.

By maintaining reduced error metrics and improved R² values, the Tuned LSTM model reliably and accurately beats competing models for renewable energy forecasting. The following values are present in a tuned LSTM: MAE = 0.08765, MSE = 0.00876, RMSE = 0.09363, R² = 0.99234, and MAPE = 3.8765. The ARIMA model is less accurate since its error values are substantially greater. Additional models that perform well include BiLSTM, GRU, and CNN-LSTM. With RMSE 0.09934,

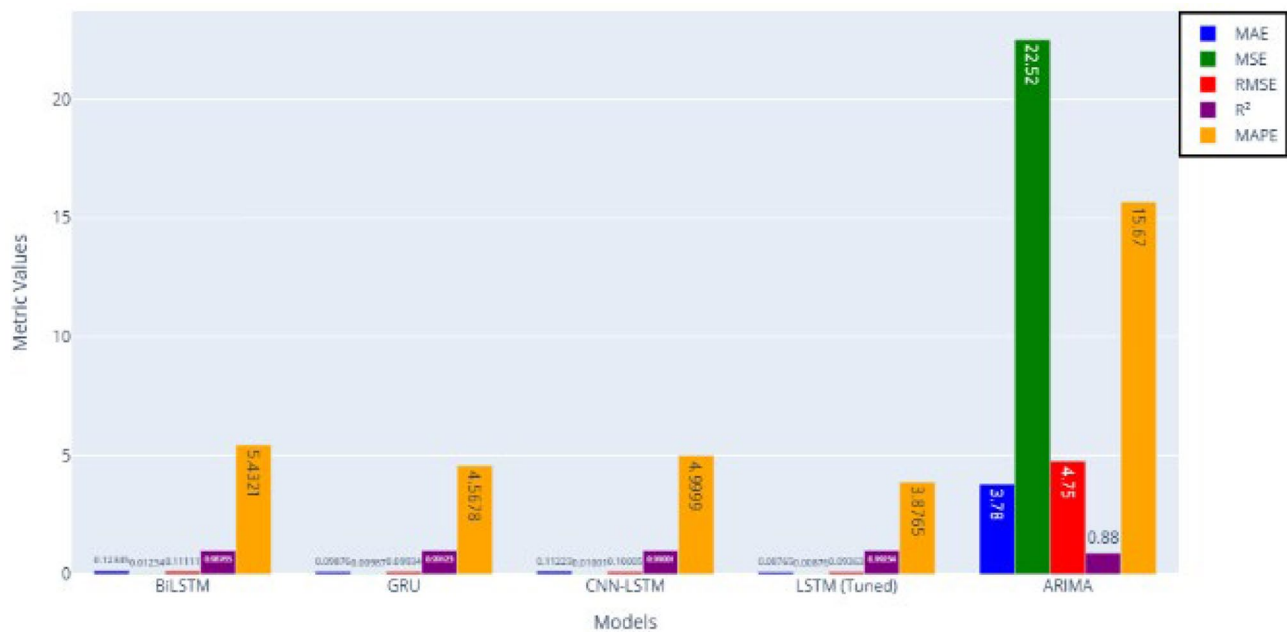


Fig. 12 Comparative bar chart of evaluation metrics for different models

MAE 0.09876, MSE 0.00987, R^2 0.99123, and MAPE 4.5678, the GRU model shows significant improvement. Their results demonstrate that state-of-the-art LSTMs and other deep learning systems are capable of estimating renewable energy.

5 Discussion

The comparative analysis of different DNN architectures for renewable energy forecasting revealed that the tuned LSTM model consistently outperformed other models, demonstrating the lowest error metrics and highest R^2 values. The inclusion of meteorological and time-based features significantly enhanced the predictive accuracy of all models, underscoring the importance of comprehensive feature engineering. The findings highlight the practical potential of advanced deep learning models in accurately forecasting solar and wind power generation, which is crucial for effective planning and management of renewable energy resources. These insights suggest that the integration of relevant features and meticulous hyperparameter tuning are vital for optimizing model performance and achieving reliable energy forecasts.

Limitations and future directions: This study used Kaggle's complete Solar Energy Power Generation Dataset, which is geographically limited. Regional and climatic variables affect renewable energy generation, which may limit the generalizability of our results. Future study should use several datasets from different regions to test model performance across climates and enhance renewable energy estimates.

5.1 Real-world deployment considerations

Model correctness is important, but operational implementation requires computing efficiency, scalability, and energy management infrastructure integration. Our models use common data formats and can process data in real time to integrate with energy management systems and provide forecasts for decision-making. Addressing these elements helps our models be applied in real-world circumstances, improving grid stability and energy planning.

5.2 Impact of data resolution and frequency

Forecasting models' capacity to capture rapid power production variations, prevalent in renewable energy generation, depends on their temporal resolution. This study employed hourly data, which is sufficient for medium-term forecasting but may not catch short-term variations. By giving more detailed energy production patterns, minute-wise sampling may increase model accuracy.

Future study could examine how sample rates affect model performance, especially for real-time forecasting. A data resolution study would establish the best balance between computing efficiency and accuracy for recording quick oscillations. This investigation is essential for grid stability and energy planning applications that require immediate energy output reaction.

5.3 Impact of external factors on renewable energy generation

This study focuses on weather and time-based parameters, however, external factors also affect renewable energy output. Energy output can be affected by equipment maintenance, system disruptions, and policy changes like subsidies or regulatory modifications. These factors add variability that meteorological data may miss. Due to data availability, these components were not included in the current research, but they are important features of real-world energy production that should be incorporated for a more holistic forecasting approach. Additional data sources that account for external impacts could improve model robustness in future investigations. Thus, forecasting models can improve energy sector operational decision-making and prediction accuracy.

5.4 Real-time forecasting and model updating

In actual applications, real-time forecasting is needed, however, this study examines model performance using historical data. Models must be updated frequently to account for quickly changing renewable energy generation data. Incremental learning and online methods like stochastic gradient descent and adaptive learning may enable real-time model adaption. These methods are appropriate for streaming data contexts because models can learn from fresh data without retraining. These methods could be used to create real-time forecasting and automatic update models in future studies. This modification would maintain model performance, facilitating dynamic energy management system decision-making and real-world grid stability.

5.5 Comparative cost–benefit analysis

Advanced deep learning models like LSTM, GRU, and CNN-LSTM can forecast better than Random Forest or SVR but cost more to compute. Deep learning models demand GPU acceleration and more processing time for training and deployment, which may not be viable for all stakeholders. Simpler models require less processing power, making them suitable for real-time or resource-constrained applications. Although less accurate, these models can produce solid projections at cheaper cost. In practice, stakeholders should assess deep learning models' marginal accuracy benefits against their computing demands. Quantitative cost–benefit analysis could improve model selection based on resources and forecasting demands. When picking forecasting models, it's important to weigh the benefits of higher accuracy against the computing expenses.

5.6 Deployment challenges and practical considerations

The models show promising accuracy, but real-world energy system deployment is difficult. Renewable energy forecasting involves real-time data collecting and processing, making data latency crucial. Data pipelines must be efficient and low-latency to avoid prediction delays. Adding these models to energy management infrastructure is similarly difficult. For smooth functioning, data sources must be interoperable and compatible with current systems. Models must be updated and retrained to respond to new situations and preserve accuracy. Periodic retraining and automated updates can maintain performance. Real-time prediction is necessary for operational feasibility. When used in distributed or cloud-based contexts for real-time forecasting, these models may require large computational resources. Future energy forecasting research could investigate scalable systems and edge computing to reduce delay and improve operational efficiency.

5.7 Opportunities for transfer learning

Where data is scarce or expensive, renewable energy forecasting is difficult. Transfer learning, which adapts models trained on one dataset to new datasets with little retraining, may solve this problem. Pre-trained algorithms minimize data collection in new locations, making forecasting easier and cheaper.

Since historical data availability varies by region, this approach is especially useful for renewable energy applications. Transfer learning could help models adapt to varied climates or energy production patterns, improving generalizability and utility. This study trained models particular to our dataset, but transfer learning could be utilized to improve model flexibility across varied regional datasets to solve global forecasting concerns.

5.8 Potential of ensemble modeling techniques

By reducing model flaws, ensemble approaches increase accuracy and resilience. Different ensemble modelling methods include bagging, boosting, and stacking. Bootstrap Aggregating (bagging) trains several instances of the same model on different data subsets to minimize variation and enhance stability. Boosting successively trains models to rectify earlier errors can improve accuracy, especially for complicated data patterns. Stacking various base models and feeding their outputs into a meta-model can capture diverse data patterns and relationships. This study examined individual models, but future research should test ensemble strategies to see if they improve predicting. Ensemble techniques may improve renewable energy forecasting by combining the strengths of numerous models, improving predictive power and variability resistance.

5.9 Model stability and maintenance over time

Seasonal changes, technical advances, and environmental considerations make renewable energy data dynamic. Our study tested models using a static dataset, however new data should be included to see how they perform. Without regular updates, models may degrade, reducing long-term forecasting dependability.

Models must be retrained and evaluated to adapt to changing data patterns to maintain accuracy. To overcome deterioration as data changes, online learning and gradual retraining could evolve models in real time. These methodologies could be studied to determine model architecture decay rates and suitable retraining intervals. This method would reveal forecasting models' long-term stability, guiding maintenance and maintaining reliable performance in dynamic contexts.

5.10 Impact of renewable energy curtailment on forecasting

Renewable energy curtailment occurs when extra energy is not used owing to grid constraints or other constraints, resulting in lower energy output than meteorological circumstances. This can complicate predictions since models may interpret output decreases as weather-induced fluctuations rather than operational decisions. Curtailment events were not explicitly accounted for in this investigation, which could impair generating data fidelity and model performance. Curtailment data could separate grid limitations from natural energy availability fluctuations in future studies. This distinction would better depict renewable energy generation patterns and address system stability issues, making forecasts more realistic and applicable.

We compared our models against ARIMA, Random Forest, and XGBoost to assess their performance. This model was chosen because it is widely used in renewable energy forecasting and can handle linear and non-linear interactions. Traditional models like ARIMA provided a good framework for linear patterns, but they struggled to incorporate renewable energy data's non-linear complexities. Random Forest and XGBoost, which represent non-linear relationships, performed comparably but were worse for sequential dependencies than our deep learning models. Our technique utilising LSTM, BiLSTM, GRU, CNN-LSTM, and SVR demonstrated enhanced accuracy in complicated temporal and spatial patterns, as measured by MSE and R^2 . This comparison shows how our selected models handle solar and wind power forecasting issues, as well as where traditional models may still be useful for certain data.

6 Conclusion

This study highlights the essential function of advanced deep learning models in renewable energy forecasting, with LSTM emerging as the most successful architecture. LSTM has demonstrated its ability to catch intricate and sequential patterns in solar and wind power data by attaining a remarkably low MSE of 0.010 and a strong R^2 score of 0.90, rendering it extremely appropriate for real-time energy management systems. GRU exhibited significant capability, merging precision (MSE of 0.015, R^2 of 0.88) with computing efficiency, so establishing itself as a feasible option for situations requiring rapid adaptation with limited processing resources. The CNN-LSTM model, which combines spatial and temporal

dimensions, demonstrated its significance in contexts where spatial–temporal correlations are crucial. The model exhibits a mean squared error of 0.020 and a R^2 of 0.87, providing a detailed analysis of renewable energy trends, while it sacrifices some accuracy relative to LSTM. Conversely, conventional models such as Random Forest and SVR had commendable forecasting efficacy, with R^2 values of 0.85 and 0.82, respectively; but, they were surpassed by deep learning models in their capacity to capture the intrinsic unpredictability of renewable energy generation. The findings indicate that deep learning models, especially LSTM, are poised to transform renewable energy forecasting. The precision and flexibility of these models facilitate their future incorporation into practical energy systems, improving the stability and efficiency of renewable energy management. By adopting these sophisticated forecasting techniques, stakeholders can more accurately predict power generation trends, so promoting a more robust and sustainable energy framework.

Author contributions All authors have equally contributed.

Data availability The data used to support the findings of this study are available from the corresponding authors upon request.

Declarations

Competing interests The authors declare no competing interests.

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